

AI-Driven Multidisciplinary Design Optimization

AIAA MDO TC Workshops on Multifidelity Methods

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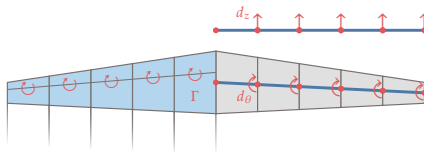
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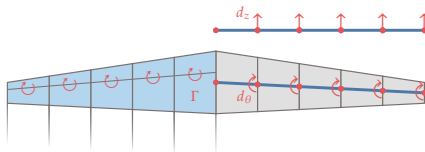
Multidisciplinary Design Optimization (1/2)

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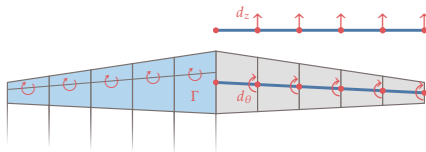


The designer wishes to minimize drag D subject to a stress constraint σ .

$$\begin{aligned} \min_{\mathbf{x}} \quad & D(\mathbf{x}; \hat{\Gamma}) \\ \text{subject to} \quad & \sigma(\mathbf{x}; \hat{\mathbf{d}}) - \sigma_{\max} \leq 0 \end{aligned}$$

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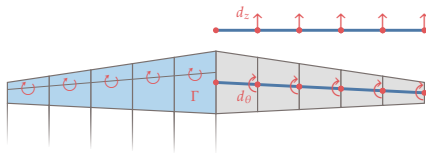
$$\text{subject to} \quad \sigma(\mathbf{x}; \hat{\mathbf{d}}) - \sigma_{\max} \leq 0$$

$$\text{while solving} \quad r_1(\Gamma; \hat{\mathbf{d}}, \mathbf{x}) = A(\hat{\mathbf{d}})\Gamma - \mathbf{v}(\hat{\mathbf{d}}) = \mathbf{0}$$

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The objective and constraint functions depend on two parameters $\hat{\Gamma}$ and $\hat{\mathbf{d}}$ which are computed using disciplinary analyses r_1 and r_2 .

Multidisciplinary Design Optimization (2/2)

Γ and $\mathbf{d}$ represent the aerodynamic vortices and structural displacements, respectively.

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$r_1(\Gamma; \hat{\mathbf{d}}) = \mathbf{0}$ and $r_2(\mathbf{d}; \hat{\Gamma}) = \mathbf{0}$ could involve computational fluid dynamics (CFD) and finite element analysis (FEA) softwares, respectively.

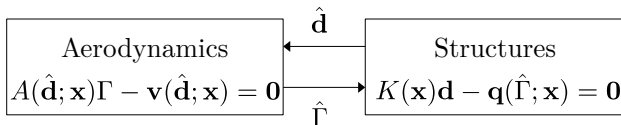
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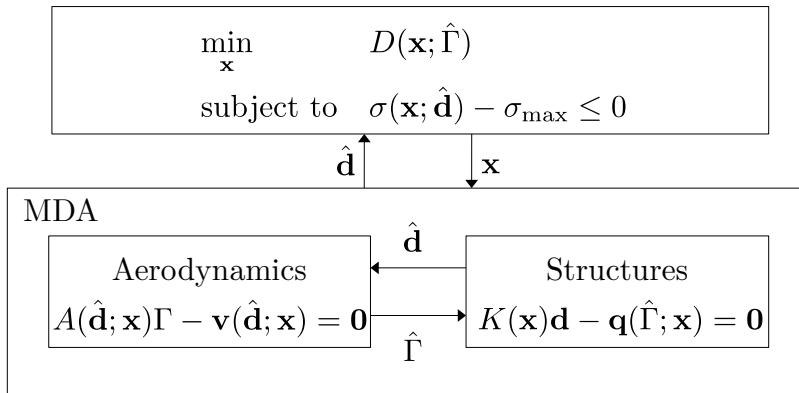
$r_1(\Gamma; \hat{\mathbf{d}}) = \mathbf{0}$ and $r_2(\mathbf{d}; \hat{\Gamma}) = \mathbf{0}$ could involve computational fluid dynamics (CFD) and finite element analysis (FEA) softwares, respectively.

A multidisciplinary analysis (MDA) is used to find $\hat{\Gamma}$ and $\hat{\mathbf{d}}$ that satisfy both r_1 and r_2 for a given wing design $\mathbf{x}$. $\hat{\Gamma}$ and $\hat{\mathbf{d}}$ are generally defined as *coupling variables*.



Multidisciplinary feasible (MDF) optimization

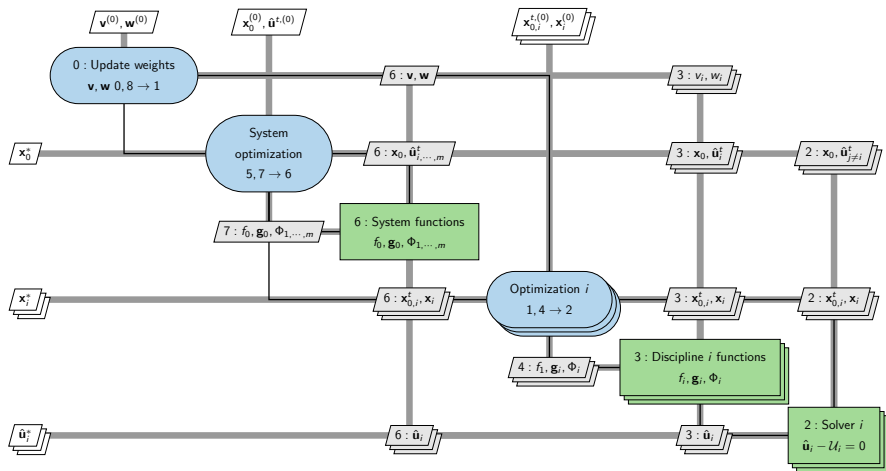
We treat the converged MDA as a blackbox that simply returns the coupling variables as a function of the design variables $\hat{\mathbf{d}}(\mathbf{x})$ and attempt to perform optimization using an algorithm of our choice.



Block diagram of MDF

Analytical target cascading (ATC)

We represent the previous ATC formulation using the following XDMS.



XDMS of ATC formulation

Non-hierarchical analytical target cascading (NHATC) (1/3)

- We follow the same exact procedure as with ATC but instead of assigning objectives and constraints to system and subsystem problems we only do so for subsystem problems.

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- The general *subproblem* i (usually one for each $i \in \{1, \dots, m\}$):

$$\begin{aligned}
 &\underset{\mathbf{x}_{0,i}^t, \mathbf{x}_i, \hat{\mathbf{u}}_{j \neq i,i}^t}{\text{minimize}} && f_i(\mathbf{x}_{0,i}^t, \mathbf{x}_i, \hat{\mathbf{u}}_{j \neq i,i}^t; \hat{\mathbf{u}}_i) + \Phi_i(\hat{\mathbf{u}}_{j \neq i,i}^t - \hat{\mathbf{u}}_{j \neq i}) + \\
 &&& \sum_{j=1, j \neq i}^m \Phi_i(\mathbf{x}_{0,i}^t - \mathbf{x}_{0,j \neq i}^t, \hat{\mathbf{u}}_{i,j}^t - \hat{\mathbf{u}}_i) \\
 &\text{subject to} && \mathbf{g}_i(\mathbf{x}_{0,i}^t, \mathbf{x}_i, \hat{\mathbf{u}}_{j \neq i,i}^t; \hat{\mathbf{u}}_i) \leq \mathbf{0} \\
 &\text{while solving} && r_i(\hat{\mathbf{u}}_i; \mathbf{x}_{0,i}^t, \mathbf{x}_i, \hat{\mathbf{u}}_{j \neq i,i}^t) = \mathbf{0}, \text{ for } \hat{\mathbf{u}}_i
 \end{aligned}$$

Non-hierarchical analytical target cascading (NHATC) (2/3)

This means that for subproblem $i \in \{1, \dots, m\}$:

- we formulate $j = m - 1$ consistency constraints between *non-local target coupling variables* $\hat{\mathbf{u}}_{i,j}^t$ and *local coupling variables* $\hat{\mathbf{u}}_i$ as $\mathbf{h}_j^c = \hat{\mathbf{u}}_{i,j}^t - \hat{\mathbf{u}}_i = \mathbf{0}$

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- we formulate $j = m - 1$ consistency constraints between *local target shared variables* $\mathbf{x}_{0,i}^t$ and *non-local target shared variables* $\mathbf{x}_{0,i}^t$ as $\mathbf{h}_{0,j}^c = \mathbf{x}_{0,i}^t - \mathbf{x}_{0,j}^t = \mathbf{0}$

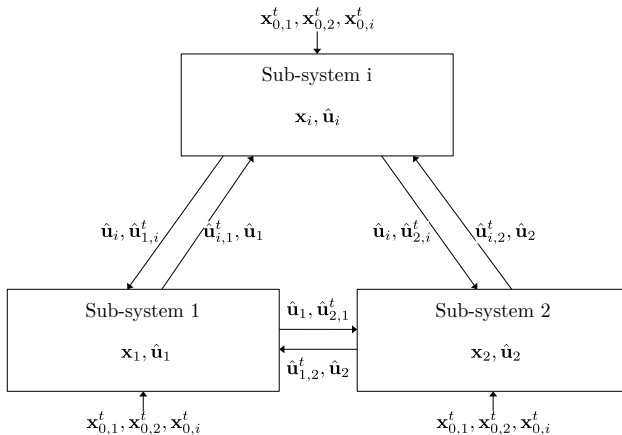
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- All subproblems are solved iteratively while updating penalty weights using ADMM coordination.

NHATC: Block diagram (3/3)

The previous NHATC formulation is shown in the following block diagram.

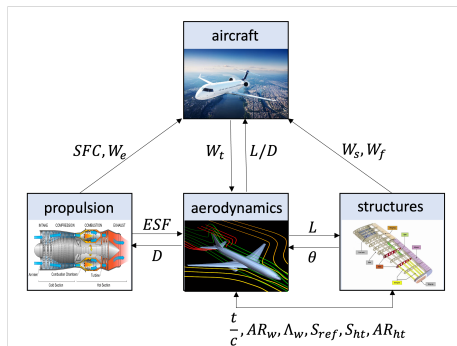


Block diagram of ATC formulation

MDO process Challenges

Key characteristics:

- Coupled physics / simulation models
- Large design spaces
- Conflicting objectives & constraints
- High computational cost



Blackbox Optimization Challenges

We consider blackbox optimization problems:

$$\min_{x \in \Omega} f(x)$$

where evaluations of f and the functions defining Ω are usually the result of a computer code (a blackbox).



BBO challenges



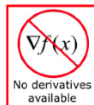
Long runtime



Large memory requirement



Software might fail



No derivatives available



Local optima



Non-smooth, noisy

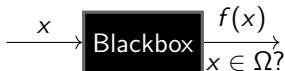
Possible bad properties of a blackbox from *J. Simonis, 2009, The Boeing Company*

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BBO challenges

The AI Opportunity

Machine learning can dramatically reduce evaluation cost, enable global search, and learn cross-disciplinary couplings from data.

AI use in Classical MDO Architectures

Architecture	Coupling strategy	Typical use	AI use
MDF	Fixed-point iteration	Tightly coupled systems	Surrogate modeling
IDF	Consistency constraints	Parallel disciplines	Bayesian optimization
AAO	All-at-once	Gradient-rich problems	Differentiable surrogates
CSSO	Coordination	Large organizations	RNNs and GRUs
BLISS	Sensitivity-based	Hierarchical systems	Sensitivity learning

Table: Common MDO architectures, their trade-offs and AI use as surrogate models.

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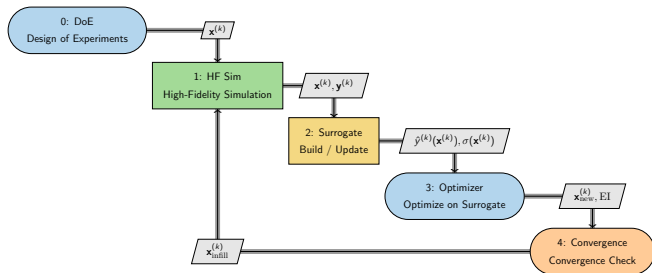
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Surrogate-Based Optimization (SBO) Workflow



Key steps: Sample $\rightarrow$ Simulate $\rightarrow$ Surrogate $\rightarrow$ Optimize $\rightarrow$ Check convergence

- SBO is a powerful framework for expensive MDO problems.
- Surrogate quality and infill strategy are critical.
- Key metric: **Expected Improvement (EI)** or **Probability of Feasibility**
- Infill criterion balances *exploitation* and *exploration*

Gaussian Process Surrogate — Quick Recap

A GP defines a distribution over functions:

$$f(\mathbf{x}) \sim \mathcal{GP}(\mu(\mathbf{x}), k(\mathbf{x}, \mathbf{x}'))$$

Prediction

$$\begin{aligned}\mu^* &= \mathbf{k}_*^\top \mathbf{K}^{-1} \mathbf{y} \\ \sigma^{*2} &= k_{**} - \mathbf{k}_*^\top \mathbf{K}^{-1} \mathbf{k}_*\end{aligned}$$

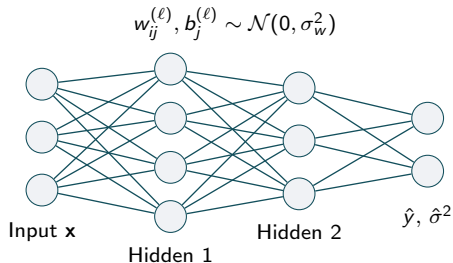
Common kernels

- Squared Exponential (RBF)
- Matérn 5/2
- Rational Quadratic

Key insight

σ^* gives a **free uncertainty estimate** — essential for active learning in MDO.

Bayesian Neural Network (BNN): Architecture and Probabilistic Prediction



$$p(y | \mathbf{x}, \mathcal{D}) = \int p(y | \mathbf{x}, \boldsymbol{\theta}) p(\boldsymbol{\theta} | \mathcal{D}) d\boldsymbol{\theta}$$

$$p(y | \mathbf{x}, \mathcal{D}) \approx \frac{1}{S} \sum_{s=1}^S p(y | \mathbf{x}, \boldsymbol{\theta}^{(s)}), \quad \boldsymbol{\theta}^{(s)} \sim p(\boldsymbol{\theta})$$

$$P(y \leq y_{\text{req}} | \mathbf{x}, \mathcal{D}) \approx \frac{1}{S} \sum_{s=1}^S \mathbb{I}(y^{(s)} \leq y_{\text{req}})$$

Quality/criteria probability

For pass/fail criteria (e.g., stress $\leq \sigma_{\text{max}}$), BNN outputs a full predictive distribution, enabling direct risk-aware decisions.

Physics-Informed Neural Networks (PINNs)

$$\mathcal{L} = \underbrace{\mathcal{L}_{\text{data}}}_{\text{observations}} + \lambda \underbrace{\mathcal{L}_{\text{PDE}}}_{\text{residual}} + \mu \underbrace{\mathcal{L}_{\text{BC}}}_{\text{boundaries}}$$

- Embed governing equations directly in the loss function
- Works with sparse / noisy data
- Suitable for **inverse problems** & sensitivity analysis
- Libraries: DeepXDE, Modulus (NVIDIA), JAX

MDO advantage

PINNs can serve as differentiable discipline models, enabling gradient propagation across the entire coupled system.

Agentic AI in MDO (1/4)

What is an agentic workflow?

- LLM-based planner decomposes design goals into tasks
- Tool-using agents call solvers, scripts, and databases
- Memory stores prior runs, constraints, and failures
- Verifier agent checks feasibility before promotion

Closed-loop cycle

Plan → Simulate → Learn → Propose → Verify

- Human sets objectives, safety limits, and trust gates
- Agent proposes candidates and justifies trade-offs

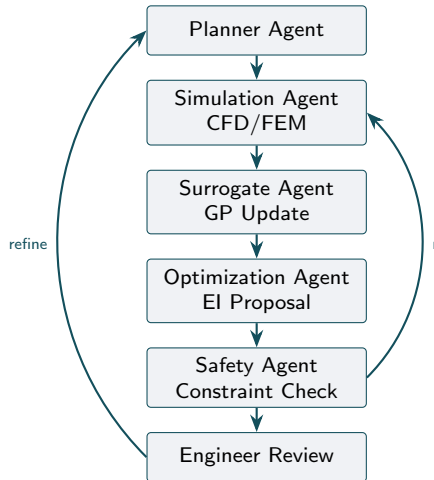
Why it matters for MDO

Agentic AI can orchestrate multi-disciplinary pipelines, reduce manual iteration overhead, and accelerate convergence while keeping engineers in the loop.

Agentic AI in MDO: Illustrative Example (2/4)

Scenario: Wing drag minimization with lift and stress constraints

- 1 **Planner agent** decomposes goals and constraints
- 2 **Simulation agent** runs CFD/FEM on candidates
- 3 **Surrogate agent** updates GP mean and uncertainty
- 4 **Optimization agent** selects EI-driven infill points
- 5 **Safety agent** filters infeasible proposals
- 6 **Reviewer agent** sends ranked options to engineer



Related references: Yao et al., *ReAct* (ICLR 2023); Wu et al., *AutoGen* (2023); Martins & Ning, *Engineering Design*

Agentic AI in MDO: Illustrative Example - Wing MDO (3/4)

$$\begin{aligned}
 \min_{\mathbf{x}} \quad & f(\mathbf{x}) = C_d(\mathbf{x}) \\
 \text{s.t.} \quad & g_1(\mathbf{x}) = |C_l(\mathbf{x}) - 1.2| - 0.1 \leq 0, \\
 & g_2(\mathbf{x}) = \sigma(\mathbf{x}) - 800 \leq 0, \\
 & \mathbf{x}^L \leq \mathbf{x} \leq \mathbf{x}^U
 \end{aligned}$$

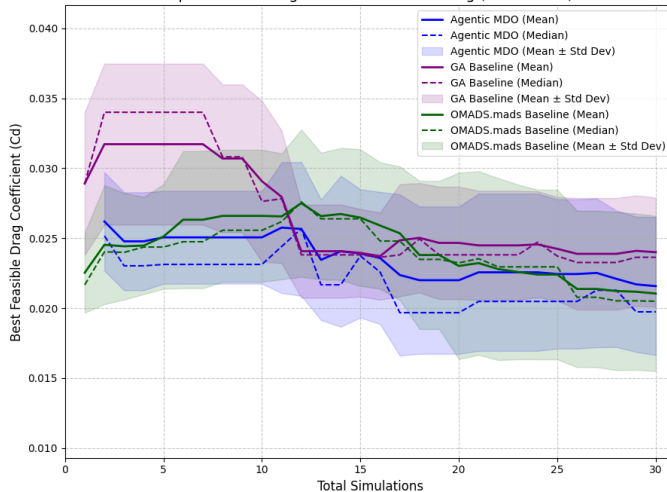
$$\mathbf{x} = [\text{AR}, \text{TR}, \text{tc}]^\top, \quad \mathbf{x}^L = [5.0, 0.2, 0.08]^\top, \quad \mathbf{x}^U = [15.0, 0.8, 0.15]^\top$$

$$C_d(\mathbf{x}) = 0.02 + \frac{0.01}{\text{AR}} + 0.05 \text{tc}^2 + 0.01(1 - \text{TR})$$

$$C_l(\mathbf{x}) = 0.5 + 0.05 \text{AR} + 0.1 \text{TR} - 0.2 \text{tc}, \quad \sigma(\mathbf{x}) = 100 \text{tc}^{-1} + 5 \text{AR} + 20(1 - \text{TR})$$

Agentic AI in MDO: Illustrative Example - Wing MDO (4/4)

Optimization Progress: Best Feasible Drag (N=20 Runs)



LLM Engagement: Orchestration Layer

Planner Agent

- Orchestrates phases: sampling, optimization, and review
- Issues goals to numerical agents (CFD/FEM/surrogate)
- Reports progress and explains next-step rationale

Reviewer Agent

- Interprets feasible candidate designs
- Ranks alternatives by objective value and risk
- Produces concise engineer-facing recommendations

Division of Labor

The LLM handles strategy, coordination, and communication; numerical solvers and optimizers handle heavy computation.

Bayesian Deep-Ensemble — Architecture

Network structure (per member)

- Two shared hidden layers with ReLU activations
- **Dual output heads:**
 - $\hat{\mu}(\mathbf{x})$ — predicted mean
 - $\hat{\sigma}^2(\mathbf{x}) = e^{\widehat{\log \sigma^2}(\mathbf{x})}$ — predicted variance
- Trained by minimising the heteroscedastic NLL loss:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N \frac{1}{2} \left[\log \hat{\sigma}^2(\mathbf{x}_i) + \frac{(y_i - \hat{\mu}(\mathbf{x}_i))^2}{\hat{\sigma}^2(\mathbf{x}_i)} \right]$$

Ensemble aggregation (M members)

Mixture mean:

$$\bar{\mu} = \frac{1}{M} \sum_{m=1}^M \hat{\mu}_m$$

Mixture variance (law of total variance):

$$\bar{\sigma}^2 = \underbrace{\frac{1}{M} \sum_m \hat{\sigma}_m^2}_{\text{aleatoric}} + \underbrace{\frac{1}{M} \sum_m \hat{\mu}_m^2 - \bar{\mu}^2}_{\text{epistemic}}$$

Models Training & Retraining Schedule

Lifecycle

- 1 Run **initial direct simulations** (default: 20) — no surrogate yet.
- 2 Once the budget is reached, fit the ensemble on all simulated data.
- 3 After every **retrain interval** of new direct simulations (default: 5), refit the ensemble with the enlarged dataset.

Input normalisation

$$\tilde{\mathbf{x}} = \frac{\mathbf{x} - \mu_x}{\sigma_x}$$

Data used for training

- Only *simulated* (true-physics) evaluations enter the training set — predicted evaluations are **never** used to retrain.
- Each ensemble member is seeded differently ($\text{seed} = 1000 + \text{member_index}$) to encourage diversity.

Warm-start

If a prior evaluation CSV exists, the surrogate is bootstrapped from it — no wasted simulations across restarts.

Prediction Pipeline — From Input to Probabilities

① Forward pass through each ensemble member

$$\hat{\mu}_m(\mathbf{x}), \quad \hat{\sigma}_m^2(\mathbf{x}) = e^{\widehat{\log \sigma^2}_m(\mathbf{x})}, \quad m = 1, \dots, M$$

② Aggregate into a single Gaussian predictive distribution $\mathcal{N}(\bar{\mu}, \bar{\sigma}^2)$ using the mixture equations (Frame 1).

③ Monte-Carlo sampling — draw $S = 2000$ samples $\{y^{(s)}\}_{s=1}^S \sim \mathcal{N}(\bar{\mu}, \bar{\sigma}^2)$ for each output (objective + constraint metrics).

④ Estimate probabilities

Probability of infeasibility

$$\hat{P}_{\text{inf}} = \frac{1}{S} \sum_{s=1}^S \mathbf{1}[\exists j : c_j^{(s)} > 0]$$

where $c_j^{(s)}$ is the j -th constraint

Probability of no improvement

$$\hat{P}_{\text{ni}} = \frac{1}{S} \sum_{s=1}^S \mathbf{1}[f^{(s)} \geq f^* - \varepsilon]$$

where f^* is the best feasible objective found so far and ε is the

Prediction Acceptance — When to Trust the AI Model?

Acceptance rule

An AI model prediction is *accepted* (i.e. the true simulator is **skipped**) if and only if:

$$\hat{P}_{\text{inf}} \geq \tau \quad \text{OR} \quad \hat{P}_{\text{ni}} \geq \tau$$

Intuition

- High $\hat{P}_{\text{inf}}$: the AI model is confident the point is infeasible — no need to waste a simulation on it.
- High $\hat{P}_{\text{ni}}$: the AI model is confident the point won't improve the best known objective — safe to skip

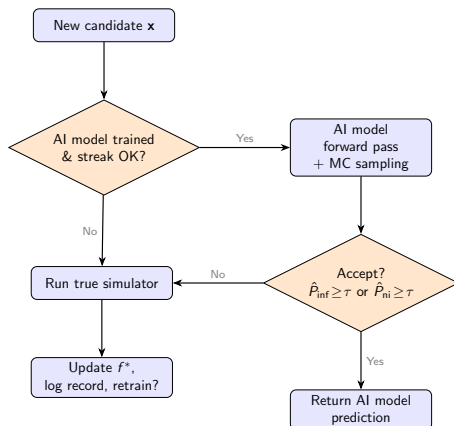
Threshold levels τ

Level	τ	Character
conservative	0.85	Cautious
average	0.65	Balanced
predictive	0.50	Aggressive

If prediction is rejected:

- 1 Run the true simulator.
- 2 Update f^* if feasible & improving.
- 3 Log $\hat{P}_{\text{inf}}, \hat{P}_{\text{ni}}$ alongside the true result for post-analysis.
- 4 Trigger retraining if the interval is reached.

Overall Evaluation Flow



Simulated evaluations

Always logged; used for training and f^* tracking.

Predicted evaluations

Logged with $\hat{P}_{inf}$ and $\hat{P}_{ni}$; **never** used to retrain the surrogate.

Tree of Parzen Estimators — Core Idea

Setup

Given a set of evaluated candidates, split them into two groups using a quantile threshold γ :

$$\mathbf{x} \in \begin{cases} \ell(\mathbf{x}) & \text{if } f(\mathbf{x}) < f^{(\gamma)} & \text{(good)} \\ g(\mathbf{x}) & \text{if } f(\mathbf{x}) \geq f^{(\gamma)} & \text{(bad)} \end{cases}$$

where $f^{(\gamma)}$ is the γ -quantile of observed objective values (typically $\gamma = 0.15$).

Key insight — Expected Improvement proxy

Why the ratio works

By Bayes' theorem:

$$p(f < f^{(\gamma)} \mid \mathbf{x}) = \frac{\ell(\mathbf{x}) \pi(\mathbf{x})}{g(\mathbf{x}) \pi(\mathbf{x})} \cdot \gamma$$

Maximising ℓ/g finds regions that are **dense under good observations** and **sparse under bad ones**.

Connection to surrogate predictions

- $\ell(\mathbf{x}) \leftrightarrow$ *predicted* distribution — where the surrogate says promising candidates live.

Methodology applied to Wing Design Problems

AI model targets and uncertainty:

$$\mathbf{y}(\mathbf{x}) = [C_d, C_l, \sigma]^\top, \quad \hat{\mathbf{y}}^{(k)} \sim \mathcal{N}(\boldsymbol{\mu}^{(k)}(\mathbf{x}), \text{diag}(\boldsymbol{\sigma}^{2,(k)}(\mathbf{x})))$$

$$\bar{\boldsymbol{\mu}}, \bar{\boldsymbol{\sigma}}^2 \quad \text{from ensemble moment matching}$$

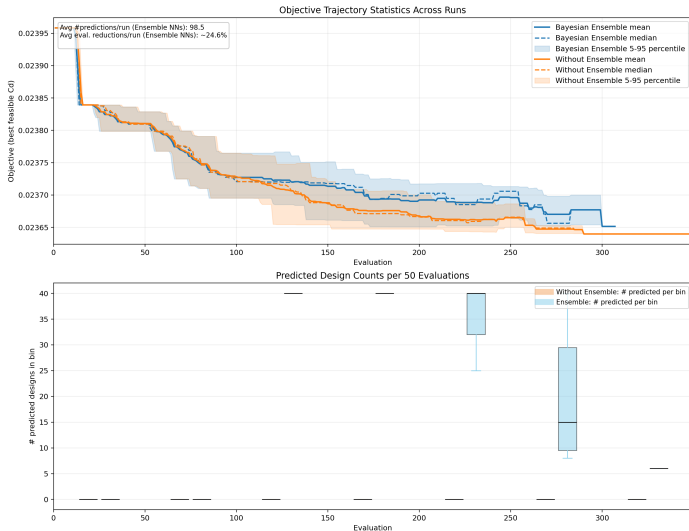
Risk metrics used in logging/decision support:

$$\hat{p}_{\text{inf}}(\mathbf{x}) = \mathbb{P}(|C_l - 1.2| > 0.1 \vee \sigma > 800)$$

$$\hat{p}_{\text{no-imp}}(\mathbf{x}) = \mathbb{P}(C_d \geq C_d^* - \delta), \quad C_d^* = \min_{\text{feasible sampled } i} C_{d,i}$$

Optimization loop (OMADS + surrogate): Initial true evaluations $\rightarrow$ train surrogate $\rightarrow$ alternate predicted/true evaluations with capped predicted streak and periodic retraining.

Results: Wing Design Validation



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- Concurrency voids in Parallel MDO
- Illustrative Example — Wing MDO

3 Case Studies

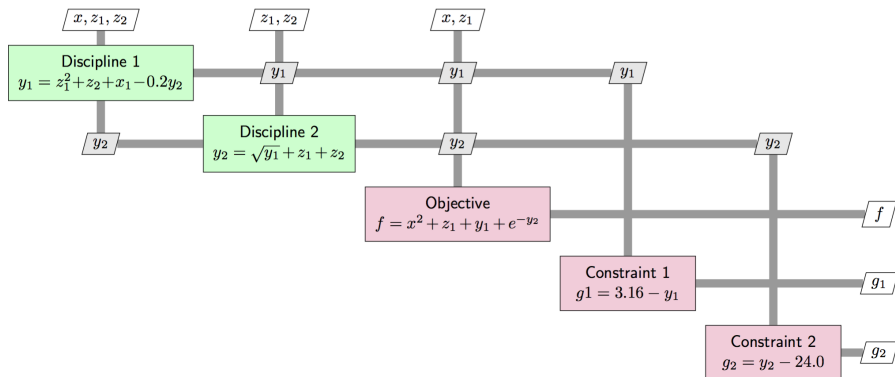
- Case Study 1 — Sellar Benchmark Problem
- Case Study 2 — Wing Airfoil Optimization
- Case Study 3 — Supersonic Business Jet Problem

4 Tools & Frameworks

- The AI tools for MDO

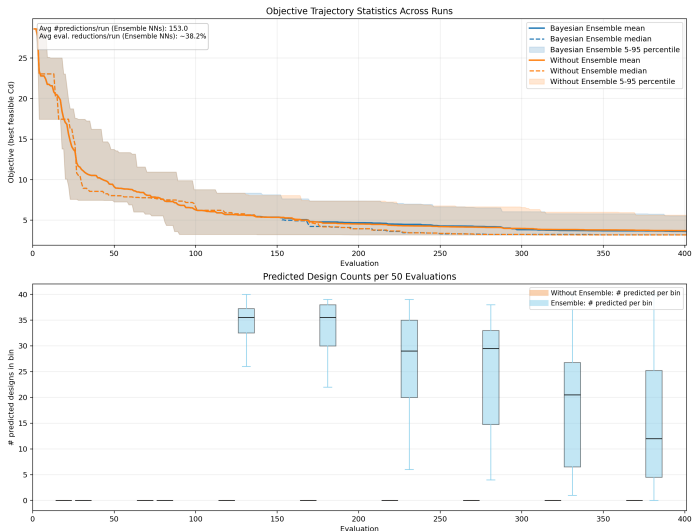
Sellar Benchmark Problem (1/2)

Purpose: compact benchmark for coupled-discipline MDO.



XDSM: optimizer with two coupled Sellar analyses and feedback exchange.

Sellar Benchmark Problem (2/2)

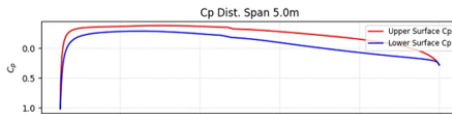
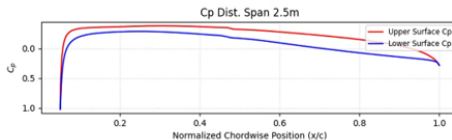
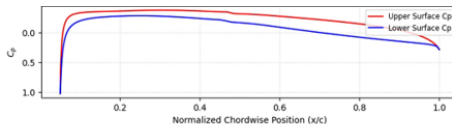
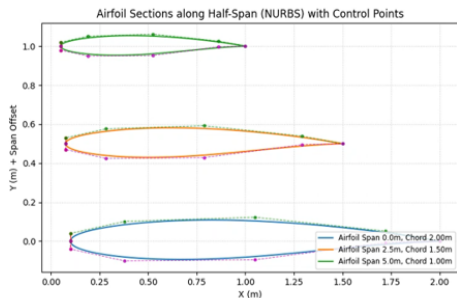


Wing Airfoil Optimization - XFOIL (1/3)

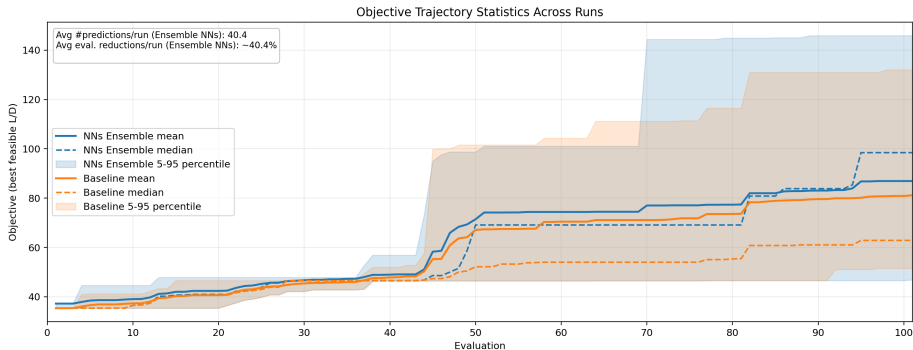
$$\begin{aligned} \min_{\mathbf{x}, \mathbf{u}_a, \mathbf{u}_s} \quad & f(\mathbf{x}) = -C_L(\mathbf{x})/C_D(\mathbf{x}, \mathbf{u}_a) \\ \text{s.t.} \quad & g_1(\mathbf{x}, \mathbf{u}_s) = \sigma_{\max}^{\text{wing}}(\mathbf{x}, \mathbf{u}_s) - \sigma_{\text{allow}} \leq 0, \\ & g_2(\mathbf{x}, \mathbf{u}_s) = C_{D_{\text{allow}}}(\mathbf{x}, \mathbf{u}_s) - C_{D_{\text{allow}}} \leq 0, \end{aligned}$$

$\mathbf{x}$: NURBS control-point perturbations; $\mathbf{u}_a$: aerodynamic state (XFOIL); $\mathbf{u}_s$: structural state (beam model).

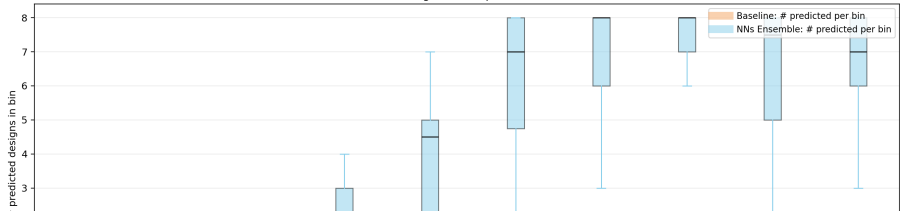
Wing Airfoil Optimization - XFOIL (2/3)



Wing Airfoil Optimization - XFOIL (3/3)



Predicted Design Counts per 10 Evaluations



Supersonic Business Jet Problem

SBJ benchmark (DMDO/NHATC): A distributed multidisciplinary optimization problem for a supersonic business jet with strong couplings and nonlinear constraints.

 **Open in Google Colab**

- **Objective:** minimize aircraft weight while satisfying mission/performance requirements.
- **Disciplines:** aircraft performance, propulsion, aerodynamics, and structures.
- **Why it matters:** a compact but realistic benchmark for coordination under coupled constraints.

Context from Notebook

The formulation coordinates subsystem targets using NHATC in a distributed MDO setting.

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- MDO Software Ecosystem

Machine learning models

- Neural Networks (BNN, FFNN, RNN)
- Kernel methods (GP, RBF)
- Tree-based models (Random Forest, XGBoost)
- Ensemble methods (stacking, boosting)

Derivative-Free Optimization

- Direct search (MADS, GPS)
- Evolutionary algorithms (GA)
- Heuristic methods (SA, PSO)
- Bayesian Optimization

Gradient-Based (Differentiable)

- Adjoint methods
- Physics-Informed Neural Nets
- Differentiable Simulators

Reinforcement Learning

- Bootstrapping
- Uncertainty-aware exploration
- Multi-agent discipline coordination

MDO Frameworks

- **DMDO** — McGill (NHATC adopted by Siemens), open-source
- **OpenMDAO** — IRT Saint-Exupéry
- **HEEDSConnect** — Siemens DISW, Commercial

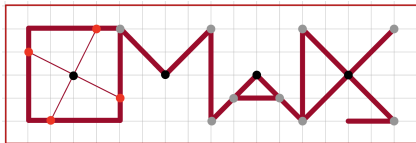
Optimization

- OMADS, Pareto MO, SO
- NSGA-II
- Ax / BoTorch (Bayesian)

Surrogate & ML

- scikit-learn (GP, RBF)
- PyTorch / TensotFlow
- SMT — Surrogate Modeling Toolbox
- DeepXDE (PINNs)

Tools for today's exercises



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Workshop Exercises Overview

#	Exercise	Tools
1	LLM in MDO	scikit-learn
2	Sellar MDO	pytorch+OMADS
3	Airfoil Optimization	XF0IL+pytorch+OMADS
4	Coupled SBJ MDO	DMD0+pytorch+OMADS

Setup

All notebooks available at:

<https://github.com/Ahmed-Bayoumy/AIAA-WS2026-AIMDO>

```
pip install omads dmdo samplersLib pytorch deepxde
```

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Where Is the Field Heading?

Near-term (1–3 years)

- Foundation models for simulation (AeroGPT, ClimateNet)
- LLM-assisted problem formulation
- Uncertainty quantification as first-class citizen
- AI-driven collaborative MDO platforms

Long-term (5–10 years)

- Fully autonomous design agents
- Real-time MDO in digital twins
- Human-in-the-loop preference learning
- Quantum-enhanced optimization

Open Challenges

- ➊ **Scalability** — curse of dimensionality for $d > 100$ design variables
- ➋ **Extrapolation** — surrogates/Al fail outside the training region
- ➌ **Certification** — regulators need explainable, verifiable AI-driven designs
- ➍ **Data scarcity** — high-fidelity runs are expensive; transfer learning is key
- ➎ **Multi-objective** — Pareto front approximation in black-box settings
- ➏ **Human factors** — bridging the gap between AI recommendations and designer intuition

Thank You!

Questions & Discussion



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`github.com/Ahmed-Bayoumy/AIAA-WS2026-AIMDO`



`https://ahmed-bayoumy.github.io/`